PERFECT MODULES STAY PERFECT UNDER FLAT BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A perfect module admits a finite resolution by finite projective modules. invoked at proof lines 19142-19144 region, stated at proof lines 19042-19044. Provides the resolution to base change.

Source reference. Original lemma 066Q: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18917–18917. Cited at proof lines 19042-19043 for the resolution of M; the conclusion is used inside the counted span.

Lemma 0.2. *Let $A \to B$ be a flat ring map. Let M be a perfect A-module. Then $M \otimes_A B$ is a perfect B-module.*

Proof. By Lemma 066Q the assumption implies that M has a finite resolution $F_\bullet$ by finite projective R-modules. As $A \to B$ is flat the complex $F_\bullet \otimes_A B$ is a finite length resolution of $M \otimes_A B$ by finite projective modules over B. Hence $M \otimes_A B$ is perfect. □

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